

TASK-1

#	Title	Steps to Reproduce	Expected vs Actual	Severity	Suspected case
1	Missing Password Strength Validation	Go to signup page and put very weak and short password.	Expected: System mandates a very strong password with specific length and characters. Actual: Accepts a weak password .	High	Absence of input validation parameters like character regex or minimum length on frontend or backend.
2	Suboptimal layout and visual hierarchy.	Open the Home page Observe the author, logos, page layout and Delete article button placement at very bottom.	Expected: A polished, modern, highly accessible user dashboard with intuitive button placement. Actual: Layout looks generic, and critical action buttons (Delete) are buried at the bottom instead of pinned near the top header actions	Low	The frontend uses raw CSS/Bootstrap skeletons without strategic UI/UX design components or user-flow optimization.
3	Duplicate Email Registration Allowed	Sign up with an email address. Log out. Try signing up a second time using the exact same email address.	Expected: App prevents submission, stating the email is already in use. Actual: The system permits multiple sign-ups under the identical email string.	Critical	The backend database schema fails to enforce a unique: true constraint on the email field, or the backend code skips a duplication check before creating the user row.
4	Missing Delete	Create a blog post.	Expected: A modal popup asking, "Are you	High	The button lack an event interceptor

	Confirmation Dialog	Click the "Delete Article" button.	<i>sure you want to permanently delete this?"</i> Actual: The post vanishes instantly upon a single click, risking accidental data loss.		(like a confirmation alert dialog) before executing the destructive API delete request.
5	Empty Content Submission on Article Edit	Open an existing article and click Edit. Wipe out the title and description completely. Click "Publish Article".	Expected: The form blocks submission, pointing out that fields are required. Actual: The system accepts and publishes a completely blank article.	High	The update/edit route lacks backend logic or frontend attributes (required) to block blank string payloads.
6	Lack of Local Image Upload Support	Go to Settings. Try to change the profile picture.	Expected: Ability to upload an image file directly from a local device storage. Actual: The app only provides a raw text input field demanding a hosted image URL.	Medium	The application is missing a file-upload controller, multi-part form data processing, and integration with a cloud storage service (like AWS S3 or Cloudinary).

Root Cause Analysis

Issue: Permissive Backend Authentication Allowing Duplicate Email Registrations

Analysis: > During testing, the web application successfully registered multiple unique user accounts using the exact same email address. This indicates a severe breakdown in the core data layer and authentication logic of the application. In a

production-ready environment, the email address serves as a primary unique identifier for user mapping, password resets, and session tokens. Allowing duplicates means that when a user attempts to log in, the database query matching the email will encounter multiple conflicting rows, likely triggering internal server crashes or logging users into the wrong accounts entirely. The root cause is a dual failure: the frontend fails to validate availability via an API check, and the backend database schema lacks a hard UNIQUE constraint on the email column.

Solution: > To fix this issue safely, the backend database schema must be modified to apply a strict unique index constraint on the user email field (e.g., email: { type: String, unique: true } in MongoDB). Additionally, the backend registration route controller must be updated to explicitly query the database for the incoming email before creating a new record, returning a clear 422 Unprocessable Entity or 400 Bad Request status code if a match already exists. Finally, the frontend user interface should catch this specific error and display an explicit, user-friendly validation warning under the email textbox.